Capping the Capacity Market: A Supply Function Equilibrium Analysis of Price Controls in PJM

March 29, 2026

Abstract

In January 2025, Pennsylvania Governor Josh Shapiro negotiated a settlement reshaping PJM's capacity market demand curve: a price cap at $325/MW-day UCAP and a price floor at $175/MW-day UCAP for the 2026/27 and 2027/28 delivery years, projected to save consumers over $21 billion. I evaluate this intervention using a calibrated Supply Function Equilibrium model of PJM's Base Residual Auction. The settlement cap is binding: under the observed market structure with three collectively pivotal suppliers, the model predicts clearing prices at the VRR Point (a) cap ($329/MW-day in 2026/27), exceeding the settlement cap by $4–$8/MW-day at the RTO level, with larger gaps in constrained Locational Deliverability Areas. The settlement suppresses prices but leaves the underlying price-cost margin unchanged. The $21 billion savings figure uses the unconstrained VRR Point (a) price as the counterfactual; a structural SFE benchmark implies a substantially smaller two-year transfer at the RTO level, and the fixed-quantity assumption overstates savings to the extent the cap depresses future entry and accelerates retirement. The complaint's lead-time argument is partial: even at short auction horizons, prices inform retirement deterrence, contracting, and forward investment expectations. Transmission expansion (increasing the Capacity Emergency Transfer Limit) reduces equilibrium prices organically, preserving the scarcity signal while addressing the structural source of binding settlement outcomes.

1 Introduction

On December 30, 2024, the Commonwealth of Pennsylvania filed a complaint with the Federal Energy Regulatory Commission alleging that PJM's capacity auction demand curve was miscalibrated, with its price cap set too high given near-term expectations for new entry. The complaint cited PJM's July 2024 Base Residual Auction for the 2025/26 delivery year, which cleared at \$269.92/MW-day, an increase of more than 800 percent from the prior year, as evidence that the existing market design imposed unnecessary costs on 65 million consumers across 13 states (Office of Governor Josh Shapiro, 2025). Six weeks later, on January 28, 2025, Governor Josh Shapiro announced an agreement with PJM in which the demand curves for the 2026/27 and 2027/28 auctions would be modified to impose a price cap of approximately \$325/MW-day (unforced capacity, UCAP) and a price floor of approximately \$175/MW-day UCAP, applied to both the RTO and all constrained Locational Deliverability Areas (Bresler and Keech, 2025). The agreement was projected to save consumers over \$21 billion over the two delivery years.

The settlement creates a two-tier price ceiling in PJM's capacity market. PJM's existing Variable Resource Requirement (VRR) curve already contains a design-level cap at Point (a), set at approximately 1.5 times the Net Cost of New Entry (CONE), serving as an investment-adequacy ceiling calibrated to the annualized cost of a reference generating unit and intended to prevent prices from permanently exceeding the entry threshold. The settlement imposes a second, lower cap at $\bar{p}_S \approx$ \$325/MW-day, below the Point (a) level, while simultaneously adding a floor at $\bar{p}_f \approx$ \$175/MW-day that clears all capacity offered at or below that level. The mechanism is implemented by reshaping the demand curve itself, giving it a flat top at $\bar{p}_S$, the existing sloped segment between $\bar{p}_S$ and $\bar{p}_f$, and a flat bottom at $\bar{p}_f$ (Bresler and Keech, 2025). This paper asks what price PJM's 2026/27 and 2027/28 auctions would have cleared at absent the settlement modification, and what the consequences of suppressing prices to $\bar{p}_S$ are.

I answer the question using a calibrated Supply Function Equilibrium (SFE) model of

the PJM Base Residual Auction. In the model, K symmetric strategic sellers simultaneously submit supply functions against PJM's VRR demand curve, which is publicly known before bidding. Following Klemperer and Meyer (1989) and Green and Newbery (1992), equilibrium is characterized by a first-order ordinary differential equation (ODE) system in the sellers' supply functions, integrated backward from the capacity-constrained boundary. Equilibrium selection imposes the Holmberg boundary condition, requiring each seller's supply function to equal installed capacity at the VRR Point (a) cap. This criterion is motivated by Holmberg (2008)'s uniqueness result for stochastic markets; in the deterministic BRA it is imposed as a selection device, and the at-cap clearing outcomes in 2026/27 and 2027/28 confirm the condition is active in practice. I calibrate the model to three benchmark delivery years using published PJM planning parameters, BRA clearing results, and IMM State of the Market cost data. The SFE equilibrium price p^* is the counterfactual price at which the unmodified VRR curve would have cleared under strategic bidding, which is what is needed to evaluate the settlement's claim of $21 billion in savings.

The remainder of the paper is organized as follows. Section 2 surveys the supply function equilibrium literature, the capacity market design literature, and the economics of price controls in oligopolistic markets. Section 3 reviews PJM's institutional context, including the VRR mechanism, auction irregularities, and the Shapiro settlement. Section 4 develops the theoretical SFE model. Section 5 describes data sources and calibration. Section 6 presents SFE equilibrium prices and comparative statics relative to the settlement cap. Section 7 analyzes the allocative consequences of the settlement's price band for entry, exit, and information. Section 8 evaluates the lead-time argument underlying the complaint. Section 9 examines the $21 billion savings claim. Section 10 considers structural alternatives to price caps. Section 11 concludes.

2 Literature Review

2.1 Supply Function Equilibria: Theory

The supply function equilibrium concept, introduced by Klemperer and Meyer (1989), considers an oligopoly in which each firm commits to a schedule mapping prices to quantities before demand is realized. Unlike Bertrand or Cournot competition, the supply function strategy allows each firm to adapt its output to whatever market price emerges after the demand shock is realized. Klemperer and Meyer show that symmetric Nash equilibria are characterized by a first-order differential equation whose solutions form a continuum between the competitive schedule and the Cournot schedule, with demand uncertainty disciplining firms to choose upward-sloping supply functions.

Green and Newbery (1992) apply the Klemperer-Meyer framework to the England and Wales electricity market, establishing the methodological template followed here: derive the symmetric ODE, integrate backward from the capacity-constrained boundary, and calibrate to observed market structure. Holmberg (2008) shows that in capacity-constrained electricity markets, the supply function equilibrium is unique when rivals' capacity constraints simultaneously bind, tying the ODE boundary condition and selecting a unique equilibrium trajectory. This paper applies that boundary condition as an equilibrium selection device in the deterministic BRA. For the asymmetric extension, Anderson and Hu (2008) provide a convergent iterative algorithm. The symmetric case applies here and I solve the ODE backward from the Holmberg boundary using the `deSolve` package in R.

2.2 Capacity Market Design

The capacity market design literature originates from the "missing money" problem: competitive energy markets, subject to regulatory price caps, suppress the scarcity rents that would otherwise finance investment in peaking generation (Joskow and Tirole, 2007; Joskow, 2008). PJM's Variable Resource Requirement curve is a downward-sloping administrative demand

schedule designed to moderate the extreme price volatility that vertical demand generates near the reliability threshold (Bowring, 2013). This paper takes the VRR design as given and analyzes how its slope governs strategic markup opportunities under concentrated supply.

2.3 Price Controls in Oligopolistic Markets

Weitzman (1974) shows that the choice between price and quantity regulatory instruments depends on the relative slopes of marginal benefit and marginal cost. When supply is inelastic in the short run, price instruments dominate. Glaeser and Luttmer (2003) demonstrate that binding price controls in imperfectly competitive markets generate misallocation qualitatively different from the standard deadweight triangle. In the SFE literature, the analysis in Section 7 draws on these results to characterize the allocative consequences of the settlement's price band. Cramton and Stoft (2005) emphasize that the capacity market's primary function is to provide a long-run investment signal. A binding cap disrupts this signal precisely when it is most informative.

3 Institutional Background

3.1 PJM's Reliability Pricing Model

PJM Interconnection operates the largest competitive wholesale electricity market in North America, coordinating more than 180,000 megawatts of generating capacity across thirteen states and the District of Columbia. PJM implemented the Reliability Pricing Model (RPM) in 2007 to address the "missing money" problem: a forward capacity market in which suppliers submit offers against a downward-sloping administrative demand schedule, the Variable Resource Requirement (VRR) curve, and the market clears at a uniform price (Bowring, 2013). The primary procurement mechanism is the Base Residual Auction (BRA).

Capacity procurement occurs separately for Locational Deliverability Areas (LDAs), geographic sub-regions constrained by transmission import limits. The Capacity Emergency

Transfer Limit (CETL) governs the maximum competitive imports available to each LDA. The panel covers the RTO and sixteen sub-LDAs across delivery years 2021/22 through 2027/28.

3.2 The Variable Resource Requirement Curve

The VRR curve is the administrative demand schedule presented to sellers in each BRA. Under the old design (delivery years 2021/22–2025/26), it is defined by three anchor points: Point (a), the price cap at $1.5 \times$ Net CONE; Point (b) at $0.75 \times$ Net CONE at the reliability requirement; and Point (c) at zero price above the reliability requirement. Net CONE is the annualized cost of a reference generating unit, net of projected energy and ancillary service revenues, computed independently for each LDA. The VRR formula is developed further in Section 4.

3.3 The 2026 VRR Redesign

FERC approved a restructured VRR curve design in 2025 (Order ER25-1357), taking effect with the 2026/27 delivery year. The redesign added a fourth anchor point creating an explicit price floor at approximately \$177–\$179 per MW-day, eliminating the collapse to zero price for surplus capacity.

Table 1 documents the BRA timeline and RTO clearing outcomes for all seven completed auctions in the sample. Both auctions conducted under the new VRR design (2026/27 and 2027/28) cleared at exactly the Point (a) price cap, motivating the SFE analysis in Sections 4–6.

Table 1: PJM Base Residual Auction Timeline and RTO Clearing Outcomes, 2021/22–2027/28

Delivery Year	BRA Date	Lead (mo.)	Clearing ($/MW-day)	Net CONE ($/MW-day)	VRR Design	At Cap
2021/22	May 2018	37.0	$140.00	$321.57	Old	No
2022/23	May 2021	12.5	$50.00	$260.50	Old	No
2023/24	July 2022	11.0	$34.13	$274.96	Old	No
2024/25	May 2024	0.8	$28.92	$293.19	Old	No
2025/26	August 2024	9.9	$269.92	$228.81	Old	No
2026/27	July 2025	10.2	$329.17	$212.14	New	**Yes**
2027/28	January 2026	16.8	$333.44	$242.52	New	**Yes**

Note: Lead time is the number of calendar months between the BRA settlement date and the start of the delivery year. RTO clearing prices and Net CONE values are in nominal dollars per MW-day. "At Cap" indicates that the RTO clearing price equaled the Point (a) price cap under the relevant VRR design. Data sources: PJM Planning Parameters; PJM BRA Results.

3.4 Auction Lead-Time Irregularity

A defining feature of the PJM capacity market during this period is the extreme irregularity of the BRA schedule. Table 1 shows that lead times compressed from 37 months (2021/22) to as little as 0.8 months (2024/25), effectively converting the 2024/25 auction into a spot procurement. The 2026/27 and 2027/28 auctions were held with 10.2 and 16.8 months of lead time, respectively, representing a partial schedule restoration. The implications of this irregularity for the price-signaling function of capacity auctions are examined in Section 8.

3.5 Market Power Mitigation: The Three Pivotal Supplier Test

PJM's primary market power mitigation mechanism is the Three Pivotal Supplier (TPS) test. Supplier i is *pivotal* in LDA ℓ if removing i's capacity would cause aggregate supply to fall

short of the reliability requirement. The Residual Supply Index for k suppliers is defined as

$$\mathrm{RSI}_k^\ell \;=\; \frac{\text{Total capacity in LDA } \ell \;-\; \text{Capacity of top-}k \text{ sellers}}{\text{Reliability requirement for LDA } \ell}. \tag{1}$$

Mitigation is triggered when $\mathrm{RSI}_1 < 1$ or $\mathrm{RSI}_3 < 1$, capping affected suppliers' offers at the Avoidable Cost Rate (ACR) (Bowring, 2013).

Table 2 summarizes TPS test outcomes across LDAs and delivery years. Three patterns stand out. First, RSI_3 falls below unity for every LDA and every delivery year, so the TPS mitigation threshold is met throughout the sample. Second, concentration is strikingly persistent (RTO RSI_3 ranges only 0.62–0.73 across six years). Third, the most extreme concentration appears in small sub-LDAs where a single seller can be individually pivotal ($\mathrm{RSI}_1 = 0$ in BGE and DEOK).

Table 2: TPS Test Results: Residual Supply Indices by LDA and Delivery Year

Delivery Year	LDA	RSI_1	RSI_3	Participants	All Pivotal
2022/23	RTO	0.81	0.73	130	Yes
2022/23	MAAC	0.69	0.37	25	Yes
2022/23	EMAAC	1.25	0.64	7	Yes
2022/23	COMED	0.43	0.36	14	Yes
2022/23	BGE	0.00	0.00	1	Yes
2022/23	DEOK	0.00	0.00	1	Yes
2023/24	RTO	0.78	0.68	134	Yes
2023/24	MAAC	0.78	0.40	11	Yes
2024/25	RTO	0.77	0.64	133	Yes
2024/25	MAAC	0.59	0.11	9	Yes
2024/25	EMAAC	0.48	0.00	2	Yes
2025/26	RTO	0.82	0.62	128	Yes
2026/27	RTO	0.82	0.64	153	Yes
2027/28	RTO	0.82	0.63	155	Yes

Note: RSI_k is the Residual Supply Index for the top-k sellers; see equation (1). "All Pivotal" indicates $RSI_1 < 1$ for each seller individually. Data source: PJM IMM State of the Market reports, various years.

3.6 The Shapiro Settlement: A Modified Demand Curve for 2026/27–2027/28

The high BRA clearing prices of 2025/26 and the anticipation of similarly elevated prices in subsequent auctions triggered a political intervention that directly modified the demand curve for two delivery years. On December 30, 2024, the Office of Governor Josh Shapiro filed a complaint against PJM's tariff with the Federal Energy Regulatory Commission under Section

206 of the Federal Power Act. The complaint's legal theory was that the VRR Point (a) price cap was set too high given lower expectations for new entry in the near term (Bresler and Keech, 2025). Because compressed auction lead times (Section 3.4) limited the ability of new generation to respond to high prices within upcoming delivery periods, the complaint argued that high prices would not produce efficient entry and instead represented an unjust and unreasonable rate. The proposed remedy was to lower the cap to $1.5 \times$ Net CONE and remove reliance on Gross CONE in the Point (a) calculation.

Following confidential settlement discussions, PJM and the Governor's office announced an agreement in principle on January 28, 2025 (Office of Governor Josh Shapiro, 2025). The settlement was presented to the PJM Markets and Operations Committee on February 7, 2025, and PJM indicated a target filing date of February 14, 2025, with the 2026/27 BRA scheduled for July 2025 and the 2027/28 BRA for December 2025 (Bresler and Keech, 2025).

Settlement terms. The agreement imposes a price band on the demand curves for both delivery years:

- **Price cap**: approximately \$325/MW-day UCAP (Unforced Capacity), equivalent to \$256.75/MW-day in ICAP terms at the assumed 79% accreditation of the dual-fuel combustion turbine reference unit.
- **Price floor**: approximately \$175/MW-day UCAP (\$138.25/MW-day ICAP), below which all available capacity clears regardless of the VRR slope.

I denote the settlement cap as $\bar{p}_S \approx$ \$325/MW-day (UCAP) and the floor as $\bar{p}_f \approx$ \$175/MW-day (UCAP) to distinguish them from the existing VRR Point (a) cap $\bar{p}$. Both values are indexed to the accreditation of the reference unit and may differ slightly from these round numbers when final ELCC values are determined.

VRR curve mechanism. The settlement does not impose an external price constraint on an otherwise unchanged auction. Instead, it modifies the shape of the VRR demand curve

itself (Bresler and Keech, 2025):

1. At prices above $\bar{p}_S$, the demand curve is flat (no capacity with offers above the cap can clear);

2. Between $\bar{p}_S$ and $\bar{p}_f$, the existing VRR sloped segment applies;

3. At prices at or below $\bar{p}_f$, the demand curve extends horizontally to clear all available capacity at the floor price.

This mechanism applies identically to the RTO demand curve and to every LDA demand curve. Clearing results continue to respect CETL transmission limitations (Bresler and Keech, 2025).

The two-cap distinction. The settlement introduces a second layer of price constraint on top of the VRR design already in place. The pre-existing Point (a) cap $\bar{p}$ is calibrated to investment incentives (approximately 1.5 × Net CONE) and has been part of PJM's market design since 2007. The settlement cap $\bar{p}_S$ is lower and is motivated by near-term consumer cost concerns rather than investment-adequacy principles. Table 1 shows that $\bar{p} = \$329.17$/MW-day in 2026/27, only \$4.17 above $\bar{p}_S = \$325$/MW-day. At the RTO level, the settlement is barely binding. The effect is largest in constrained LDAs where local Point (a) caps substantially exceeded \$325 in prior years. The analysis in Sections 6–10 compares the SFE equilibrium price p^* (computed against the unmodified VRR curve) to both caps, establishing where the settlement is and is not binding.

4 Model

4.1 Setup

The BRA is modeled as a static, complete-information game in which sellers simultaneously submit supply functions before the auction clears. This formulation follows Klemperer and

Meyer (1989) and, for the electricity market application, Green and Newbery (1992). The timing is: (1) PJM publishes the VRR curve parameters for the upcoming delivery year; (2) sellers submit supply functions; (3) PJM clears the auction at the uniform price where aggregate supply meets the VRR curve; (4) sellers deliver committed capacity during the delivery year.

Assumption 1. *There are $K \geq 2$ strategic sellers, indexed $i = 1, \ldots, K$, and a competitive fringe. Each strategic seller i holds a fixed capacity portfolio $\bar{q}_i > 0$ measured in megawatts. A supply function $S_i : [0, \bar{p}] \to [0, \bar{q}_i]$ is a nondecreasing, twice continuously differentiable mapping from price to quantity. Each seller simultaneously commits to a supply function before the auction clears.*

Assumption 2. *The competitive fringe bids at its avoidable cost rate $c_f > 0$. Fringe supply is $S_f(p) = 0$ for $p < c_f$ and $S_f(p) = \bar{Q}_f$ for $p \geq c_f$, where $\bar{Q}_f > 0$ is total fringe capacity. The fringe is price-taking and its behavior is common knowledge.*

Assumption 3. *Each strategic seller has a constant marginal cost $c > 0$ for output $q \leq \bar{q}_i$ and infinite marginal cost above capacity. For the symmetric calibration I set c equal to the sector-average avoidable cost rate published by the PJM IMM.*

The market clears at the price p^* satisfying

$$\sum_{i=1}^{K} S_i(p^*) + S_f(p^*) = D(p^*), \tag{2}$$

where $D(\cdot)$ is the VRR demand schedule described below. Every seller receives the same clearing price p^* for its committed capacity. The *residual demand* facing seller i at price p is

$$R_i(p) \;=\; D(p) - \sum_{j \neq i} S_j(p) - S_f(p). \tag{3}$$

In equilibrium, $S_i(p^*) = R_i(p^*)$. Because $D(\cdot)$ and all rivals' strategies are publicly known before the auction, there is no demand uncertainty: the residual demand curve each seller

faces is deterministic.[1] This determinism means firms observe the VRR parameters before submitting offers. The boundary condition is imposed as an equilibrium selection criterion, as discussed in Section 4.4.

4.2 The VRR Demand Schedule

PJM's Variable Resource Requirement curve is a piecewise-linear, publicly-announced downward-sloping demand schedule. Under the old design (delivery years 2021/22–2025/26) it is parameterized by three anchor points (p_a, q_a), (p_b, q_b), and $(0, q_c)$:

$$D(p) = \begin{cases} q_a & p \geq p_a \\ q_a + \dfrac{q_b - q_a}{p_b - p_a}(p - p_a) & p_b \leq p < p_a \\ q_b + \dfrac{q_c - q_b}{0 - p_b}(p - p_b) & 0 \leq p < p_b, \end{cases} \tag{4}$$

where $p_a = 1.5 \times$ Net CONE is the price cap and $p_b = 0.75 \times$ Net CONE is the reliability-requirement price. The quantities $q_a < q_b < q_c$ are the corresponding capacity levels, with q_b equal to the reliability requirement. Because $D(p) = q_a$ for all $p \geq p_a$, the VRR curve is effectively inelastic above the cap.

Under the new design (delivery years 2026/27 onward), FERC Order ER25-1357 introduces a fourth anchor point (p_f, q_f) and (p_f, q_d) with $q_f > q_b$, creating a flat floor segment at price

[1] Holmberg's 2008 uniqueness proof formally requires stochastic demand with positive probability of all capacity being demanded. In the deterministic BRA setting, the boundary condition $s(\bar{p}) = \bar{q}$ is imposed as a selection criterion motivated by that result. The at-cap clearing outcomes in 2026/27 and 2027/28 confirm the condition is active in practice.

$p_f \approx$ \$177– \$179/MW-day. The new schedule is:

$$D(p) = \begin{cases} q_a & p \geq p_a \\ q_a + \dfrac{q_b - q_a}{p_f - p_a}(p - p_a) & p_f \leq p < p_a \\ q_d & 0 \leq p < p_f. \end{cases} \tag{5}$$

The flat floor segment at p_f means that clearing prices cannot fall below p_f along the VRR, changing the elasticity profile that sellers face and thus the equilibrium markup. The comparative static in Section 6.3 isolates this design change.

The slope $D'(p)$ is piecewise constant and satisfies:

Assumption 4. *$D'(p) < 0$ on the sloped segments; $D'(p) = 0$ on the flat segments (at and above p_a, and below p_f under the new design). The function $D(\cdot)$ is known to all players before the auction.*

4.3 Firm Optimization and the SFE Condition

Seller i chooses $S_i(\cdot)$ to maximize profit. Taking rivals' supply functions as fixed, the profit at any market-clearing price p is $\pi_i = p \cdot S_i(p) - c \cdot S_i(p) = (p - c) \cdot S_i(p)$, since marginal cost is constant at c. The seller's problem is equivalent to maximizing profit along its residual demand curve $R_i(\cdot)$. At price p, if seller i considers increasing its offered quantity by $\mathrm{d}q$, the market-clearing price must fall by $\mathrm{d}p$ to preserve equation (2):

$$\mathrm{d}p \;=\; \frac{\mathrm{d}q}{D'(p) - \sum_{j \neq i} S_j'(p) - S_f'(p)}. \tag{6}$$

Since $D'(p) < 0$ and $S_j'(p) \geq 0$, the denominator is strictly negative: an increase in supply lowers the clearing price. The seller's marginal revenue from increasing quantity at price p is:

$$\mathrm{MR}_i(p) \;=\; p + \frac{S_i(p)}{D'(p) - \sum_{j \neq i} S_j'(p) - S_f'(p)}. \tag{7}$$

Setting marginal revenue equal to marginal cost c yields the *supply function equilibrium condition*:

Proposition 1. *In a supply function equilibrium, each seller i's strategy $S_i(\cdot)$ satisfies*

$$S_i(p) \;=\; -\Big(p-c\Big)\Big[D'(p) - \textstyle\sum_{j\neq i} S_j'(p) - S_f'(p)\Big] \tag{8}$$

at every price p at which the market can clear. This is a system of K coupled first-order ordinary differential equations in the supply functions $\{S_1, \ldots, S_K\}$.

Equation (8) is the K-seller generalization of the first-order condition derived by Green and Newbery (1992) for the symmetric duopoly (their equation 3) and of the general condition in Anderson and Hu (2008) (their equation 1). Its economic content is straightforward: seller i equates its residual-demand markup $(p-c)$ to the inverse of the residual demand elasticity at that price. When more sellers compete (K large) or the VRR slope is steep (large $|D'|$), the system requires a smaller markup at each price, pushing the equilibrium supply function toward the competitive schedule.

4.4 Equilibrium Selection and the Boundary Condition

The ODE system (8) generally admits a continuum of solutions. Klemperer and Meyer (1989) show that under demand uncertainty, the set of symmetric supply function equilibria is an interval bounded above by the Cournot schedule (where each firm behaves as if rivals are at capacity) and below by the competitive schedule (where $p = c$). Any trajectory between these two stationary solutions is a Nash equilibrium. This multiplicity complicates both prediction and comparative statics.

I select from this continuum using the boundary condition from the following result, which applies to the capacity-auction setting:

Proposition 2 (Holmberg 2008)**.** *Suppose sellers are symmetric, demand is inelastic above the price cap $\bar{p}$, and the probability that aggregate demand reaches the capacity ceiling is*

strictly positive. Then there exists a unique symmetric supply function equilibrium. In this equilibrium each seller's supply function satisfies the boundary condition

$$S_i(\bar{p}) \;=\; \bar{q}_i \quad \textit{for all } i. \tag{9}$$

The intuition is direct. When rivals' capacity constraints simultaneously bind, seller i faces perfectly inelastic residual demand. Its monopoly optimum at that moment is to set price equal to the cap $\bar{p}$; any lower price sacrifices revenue without reducing quantity (since residual demand is inelastic), and any higher price is infeasible. This forces the supply function to pass through the point $(\bar{p}, \bar{q}_i)$, tying the arbitrary constant in the ODE and selecting a unique trajectory.

In the PJM BRA, the price cap $\bar{p}$ equals the VRR Point (a) price, and Assumption 4 ensures $D'(\bar{p}) = 0$ (the VRR is flat at the cap). The at-cap clearing outcomes in 2026/27 and 2027/28 (Table 1) confirm that the boundary condition is active in those years, consistent with Holmberg's (2008) selection criterion: PJM's reliability target requires procuring essentially all installed capacity, making binding-capacity outcomes the empirically relevant case. I therefore integrate the ODE (8) *backward* from the boundary condition $S_i(\bar{p}) = \bar{q}_i$, following the solution method of Green and Newbery (1992).

4.5 The Symmetric Special Case

For calibration and comparative statics I begin with the symmetric case in which all K strategic sellers are identical: $\bar{q}_i = \bar{q}$ and $c_i = c$ for all i. In a symmetric Nash equilibrium, $S_i(p) = s(p)$ for all i. Substituting into equation (8):

$$s(p) \;=\; -\Big(p - c\Big)\Big[D'(p) - (K-1)\, s'(p) - S_f'(p)\Big]. \tag{10}$$

Under Assumption 2, $S_f'(p) = 0$ almost everywhere. Rearranging (10) for $s'(p)$ yields the *symmetric SFE ordinary differential equation*:

$$\boxed{s'(p) \;=\; \frac{1}{K-1}\left[D'(p) + \frac{s(p)}{p-c}\right], \quad p \in [c, \bar{p}),} \tag{11}$$

with boundary condition $s(\bar{p}) = \bar{q}$. This is a first-order, nonautonomous ODE in the single unknown function $s(\cdot)$. It is the K-firm generalization of Green and Newbery's equation (3), to which it reduces when $K = 2$ and $S_f \equiv 0$.

Equation (11) has a direct economic interpretation. The term $D'(p)$ captures the direct effect of price on quantity demanded: a steeper VRR slope (larger $|D'|$) makes $s'(p)$ more negative, meaning each firm's equilibrium supply function falls more steeply toward the cap as price rises. The term $s(p)/(p-c)$ is a markup-boundary feedback: as p approaches c from above, this term diverges and forces $s'(p)$ to remain bounded away from the demand slope, enforcing the lower boundary of the supply function and preventing equilibrium offers below cost. At prices well above c, the feedback term partially offsets $D'(p)$, making the supply function steeper in price than a linear interpolation between c and $\bar{p}$ would predict. More competitors (larger K) divide the denominator, flattening each firm's supply function toward the competitive benchmark.

The equilibrium clearing price p^* is found by solving

$$K \cdot s(p^*) + S_f(p^*) = D(p^*) \tag{12}$$

for p^* after integrating (11) backward from $\bar{p}$. The implied price-cost margin (Lerner index) is

$$L \;=\; \frac{p^* - c}{p^*}, \tag{13}$$

which measures the equilibrium price-cost margin reported in Section 6. The competitive benchmark corresponds to $c = p^*$ ($L = 0$), attained as $K \to \infty$ or as $|D'(p)| \to \infty$ (perfectly

elastic demand). The Cournot upper bound on the markup is $L^{\text{Cournot}} = 1/(K \cdot |D'(p^*) \cdot p^*/D(p^*)|)$, the standard inverse-elasticity rule evaluated at market demand (which equals the residual demand elasticity in the symmetric case).

The symmetry assumption. In reality, the three largest PJM capacity sellers hold heterogeneous portfolios: a mix of baseload nuclear, peaking gas, renewables, and demand response with widely differing cost structures and capacity sizes. Treating them as symmetric is a deliberate first-order approximation, not a statement about the actual market. The justification is tractability: the symmetric case reduces the K-dimensional system of coupled ODEs in equation (8) to the single scalar ODE (11), making the model calibratable from publicly available aggregate data. This is the standard approach in the calibrated SFE literature (Green and Newbery, 1992; Sioshansi and Oren, 2007). The direction of the asymmetry bias on the average price-cost margin is ambiguous: a lower-cost seller has a higher markup than the symmetric benchmark; a higher-cost seller has a lower one. On average, these effects partially offset. I treat the symmetric Lerner as an estimate of the capacity-weighted average markup across the strategic sellers, and note that the spread around this average is an object of interest for future asymmetric extensions.

Equation (11) is solved numerically using a fourth-order Runge–Kutta (RK4) integrator with fixed step size $h = 0.05\,\$/\text{MW-day}$. To improve stability I apply the substitution $\tau = \bar{p} - p$, which converts the backward initial-value problem into a standard forward IVP with $s(0) = \bar{q}$. The piecewise nature of $D'(p)$ is accommodated by restarting the integration at each VRR kink point: the integration grid is split at p_b (old design) or p_f (new design), so the integrator never crosses a discontinuity in $D'(p)$ within a single step. Under the new design, $D(p)$ has a jump discontinuity at p_f (from q_b approaching from above to $q_d > q_b$ below). Equilibrium is resolved by identifying p^* as the price at which aggregate supply meets the sloped-segment demand from above, which coincides with p_f when $K \cdot s(p_f) + S_f(p_f) = q_b$. Full details of the numerical implementation are provided in Section 5.

5 Calibration

5.1 Data Sources

The calibration draws on four publicly available datasets produced by PJM and the PJM Independent Market Monitor (IMM).

PJM Planning Parameters. PJM publishes the planning parameters for each BRA in advance of the auction, including the VRR curve anchor points, Net CONE, CETL, and reliability requirement for every LDA. These files are available in Excel format and provide the complete VRR demand schedule that sellers observe when submitting capacity offers. I processed the planning parameter files for delivery years 2021/22 through 2028/29 using a custom parser that extracts LDA-level parameters for each VRR design vintage and handles the three-point (old) and four-point (new) curve specifications separately. Spot-checks against published PJM summary tables confirm accuracy (e.g., 2023/24 RTO Net CONE = \$274.96 per MW-day).

PJM BRA Results. Following each auction, PJM releases clearing prices and quantities by LDA. I processed seven completed auctions (2021/22 through 2027/28). The 2028/29 BRA had not yet been conducted at the time of data collection. Clearing results are verified against PJM press releases (e.g., 2025/26 RTO clearing = \$269.92 per MW-day; 2026/27 RTO clearing = \$329.17).

PJM IMM State of the Market Reports. The PJM IMM publishes an annual State of the Market (SotM) report containing auction-by-auction summaries of market concentration, TPS test outcomes, and offer-curve characteristics. I extracted Residual Supply Index values and participant counts from the relevant report volumes using a PDF-parsing pipeline. Volume 2022 covers delivery years through 2022/23, and Volume 2025 covers delivery years 2023/24 through 2027/28. Due to layout variation across report vintages, RSI data is available for a

subset of LDAs and delivery years, as documented in Table 2. The 2021/22 delivery year predates the RSI extraction window and is excluded from the market structure analysis.

Auction Lead Times. BRA settlement dates and delivery-year start dates were coded manually from PJM auction notices and verified against the data in Table 1. Lead time is defined as the number of calendar months between the BRA settlement date and the first day of the delivery year. The highly irregular lead-time pattern documented in Section 3.4 is incorporated as a covariate in the calibration of the investment-signal environment.

The four datasets are joined at the LDA–delivery-year level to produce a master panel with 110 observations spanning 17 LDAs and seven delivery years (2021/22 through 2027/28).

5.2 VRR Curve Parameters

Table 3 reports the VRR curve parameters for the RTO under both the old and new designs for the six delivery years with complete BRA data. Under the old design, Point (a) equals $1.5\times$Net CONE and Point (b) equals $0.75\times$Net CONE by construction, so the ratio $p_a/p_b = 2$ is exact for all old-design years except 2025/26, where a transitional administrative cap produced a Point (a) price above the standard $1.5 \times$ Net CONE multiple.[2] Under the new design, Points (b) and (c) equal the explicit floor price of approximately \$177–\$179 per MW-day, and Point (a) is calibrated by FERC Order ER25-1357.[3]

[2]The 2025/26 Net CONE of \$228.81 per MW-day implies a standard cap of \$343.22, but the published Point (a) was \$451.61. This discrepancy reflects the fact that VRR anchor points are fixed at planning-parameter publication and may diverge from the final Net CONE if energy and ancillary service revenues are revised substantially between publication and auction. I use the published anchor points as the realized VRR parameters in all calibrations.

[3]The new-design Point (a) prices (\$329.17 for 2026/27, \$333.44 for 2027/28) are not exactly $1.5\times$Net CONE ($1.5 \times \$212.14 = \$318.21$ and $1.5 \times \$242.52 = \$363.78$ respectively). The \$329.17 figure reflects the settlement implementation: FERC Order ER25-1357, which implemented both the new VRR design and the settlement's cap provision, produced a Point (a) approximately \$4 above the nominal settlement cap of \$325/MW-day UCAP. The precise derivation follows the same dynamic as the 2025/26 deviation (see footnote above): VRR anchor points are fixed from planning parameters published before the auction and may incorporate accreditation adjustments that produce a Point (a) differing from a simple Net CONE multiple. I treat the published \$329.17 and \$333.44 as the effective VRR boundary conditions and note that the settlement's flat-top at \$325 creates a binding gap of \$4.17 and \$8.44 respectively below these Point (a) values.

Table 3: RTO VRR Curve Parameters, 2022/23–2027/28

Delivery Year	Net CONE ($/MW-day)	Pt. (a) Price	Pt. (a) MW	Pt. (b) Price	Pt. (b) MW	VRR Design
2022/23	$260.50	$390.75	136,076	$195.38	139,656	Old
2023/24	$274.96	$412.44	135,914	$206.22	139,473	Old
2024/25	$293.19	$439.79	138,341	$219.89	141,910	Old
2025/26	$228.81	$451.61	133,554	$171.61	137,160	Old
2026/27	$212.14	$329.17	134,887	$177.24	136,361	New
2027/28	$242.52	$333.44	141,681	$179.55	143,218	New

Note: Prices in nominal dollars per MW-day; quantities in MW. Under the old design, Point (c) is the zero-price floor at a quantity above Point (b); under the new design, Points (b) and (c) share the floor price and a flat segment extends to Point (d). The 2025/26 Point (a) price exceeds $1.5 \times$ Net CONE due to a transitional administrative cap (see footnote above). Data source: PJM Planning Parameters.

The panel encompasses 110 LDA–delivery-year cells with observed clearing prices across 17 LDAs. The three benchmark calibration years used throughout Sections 4 and 6 are selected to represent three distinct capacity market regimes: *2023/24*, a low-price year under the old VRR design exhibiting equilibrium prices consistent with competitive pricing under TPS mitigation ($34.13 per MW-day, $\text{RSI}_3 = 0.68$); *2025/26*, a high-price year under the old VRR design that preceded the redesign ($269.92 per MW-day, $\text{RSI}_3 = 0.62$); and *2026/27*, the first auction under the new VRR design, which cleared at exactly the Point (a) cap ($329.17 per MW-day, $\text{RSI}_3 = 0.64$). Together, these three years span the full range of price outcomes in the sample and allow the comparative static analysis to isolate the effect of VRR design changes from variation in market structure and cost parameters.

5.3 Market Structure

Capacity market concentration in PJM is high by any standard measure. The PJM IMM reports LDA-level Herfindahl-Hirschman Index (HHI) values in its annual State of the Market (SotM) reports. In every delivery year covered by the sample, the RTO-level HHI exceeds 2,500 and most constrained LDAs exceed 4,000, thresholds the Department of Justice classifies as highly concentrated markets under standard antitrust benchmarks.

The key calibration input is the number of strategic sellers, K. I use the Residual Supply Index with threshold three (RSI_3) as the calibration criterion: $\text{RSI}_3 < 1$ means that the three largest sellers collectively hold pivotal capacity, i.e., no competitive outcome is supportable without their participation. In all three benchmark years, the RTO-level RSI_3 falls below unity (0.68, 0.62, and 0.64 for 2023/24, 2025/26, and 2026/27, respectively), consistent with at most three large sellers being jointly pivotal. The 2026/27 at-cap clearing outcome is also consistent with $K \leq 3$: the model predicts interior clearing for four or more symmetric sellers in that year (Section 6.2), so the observed at-cap outcome aligns with a low effective strategic-seller count under the maintained model. The IMM TPS test confirms individual pivotality for all three largest sellers in every constrained LDA, supporting $K = 3$ over $K = 2$ as the baseline. I therefore set $K = 3$ as the baseline strategic count, and examine sensitivity to this assumption via the K comparative static in Section 6.2. Because RSI data is available only at the RTO level for most delivery years, I impose the same $K = 3$ structure on each LDA calibration and allow cross-LDA variation to enter through the VRR parameters and CETL-based fringe capacity.[4]

5.4 Cost Parameters

The avoidable cost rate (ACR) for each strategic seller is set to $c = \$150$ per MW-day, the combined-cycle sector average reported in the 2025 PJM IMM State of the Market (Table

[4] LDA-level TPS test outcomes reported in the IMM SotM confirm that at least three suppliers pass the pivotal test in all constrained LDAs, consistent with $K = 3$.

3-8). Combined-cycle units represent the marginal capacity technology in PJM's current resource mix, and the IMM identifies the combined-cycle ACR as the relevant benchmark for capacity-offer mitigation under the TPS test. The symmetric-firm assumption sets all $K = 3$ sellers to the same c, and heterogeneity in costs and capacity sizes is abstracted away. It is important to note that the ACR is a proxy for opportunity cost, not its direct measure. The true avoidable cost of offering a megawatt of capacity depends on projected energy-market revenues, fuel hedge positions, maintenance schedules, and the firm's financial structure, none of which are publicly observable at the portfolio level. The sector-average combined-cycle ACR therefore understates cost heterogeneity across strategic sellers. The sensitivity of equilibrium markups to this approximation is examined directly in Table 6, which varies c over \$100–\$200 per MW-day at $K = 3$ and shows that the qualitative settlement-binding finding is robust across this range.

Fringe supply is modeled as an inelastic block offered at c: any capacity not controlled by the K strategic sellers is assumed to offer at cost. At the RTO level, fringe capacity is set to total installed capacity minus the capacity attributed to the three largest sellers. This residual is scaled with γ to explore fringe-size sensitivity. For individual LDA calibrations, the Capacity Emergency Transfer Limit (CETL) serves as the effective fringe capacity, since CETL measures the maximum deliverable import from the rest of PJM into the LDA. LDA-level CETL values by delivery year are drawn directly from the PJM planning parameter files described in Section 5.1.

5.5 Benchmark Delivery Years

I select three delivery years as calibration benchmarks, each representing a distinct capacity market regime. *2023/24* is a low-price interior year (\$34.13/MW-day) under the old VRR design in which PJM's TPS mitigation was broadly applied. The gap between the model's unmitigated SFE price and the observed clearing price is consistent with an interior equilibrium of the model and illustrates the effect of offer-price mitigation. *2025/26* is a high-price year

($269.92/MW-day) that preceded the VRR redesign. It exhibits a large upward price spike relative to prior years and partial TPS application, making it useful for isolating the effect of VRR slope on equilibrium markups. *2026/27* is the first auction under the new VRR design and cleared at the Point (a) cap ($329.17/MW-day), making it the most informative benchmark year for assessing whether the maintained model is consistent with a low effective strategic-seller count.

The three benchmark years also span the range of outcomes relative to the Shapiro settlement cap of $325/MW-day (Section 3). In 2023/24, the settlement would be entirely non-binding ($p^* = \$330$ exceeds $325 but clearing occurred at $34 under mitigation; the settlement cap is irrelevant). In 2026/27, the settlement cap is binding at the RTO level ($p^* = \$329 > \$325$). This span allows the policy analysis in Sections 7–10 to evaluate both binding and non-binding settlement scenarios.

I exclude *2024/25* from the benchmark set. PJM conducted the 2024/25 BRA as a spot auction held only 0.8 months before the delivery year began, compared to the approximately 37 months of the 2021/22 auction, the most recent held near the three-year forward horizon the RPM was designed around (Table 1). This compressed timeline eliminates any forward investment signal and makes the 2024/25 clearing price ($28.92/MW-day) an unreliable guide to equilibrium capacity incentives. The SFE framework, which embeds a long-run investment signal in the VRR demand curve, is not applicable.

6 Results: Equilibrium Prices vs. the Shapiro Settlement

6.1 Baseline Equilibrium

Table 4 presents the baseline symmetric SFE results for the three benchmark delivery years, calibrated with $K = 3$ strategic sellers and $c = \$150$/MW-day (the sector-average ACR for

combined-cycle generators from the 2025 IMM State of the Market). For each year, the table reports the model-predicted equilibrium price p^*, the observed BRA clearing price, and the implied price-cost margin (Lerner index) $L = (p^* - c)/p^*$.

Table 4: Baseline SFE Equilibrium: $K = 3$, $c = \$150$/MW-day

Year	Design	$\bar{p}$	$\bar{p}_S$	p^*	p_{actual}	L	Outcome
2023/24	Old	\$412	—	\$330	\$34	0.55	Interior
2025/26	Old	\$452	—	\$426	\$270	0.65	Interior
2026/27	New	\$329	\$325	\$329	\$329	0.54	At cap

Notes: $\bar{p}$ is the VRR Point (a) price cap. $\bar{p}_S$ is the Shapiro settlement cap (\$325/MW-day UCAP), applicable to the 2026/27 and 2027/28 delivery years; "—" indicates the settlement does not apply. p^* is the model-predicted SFE clearing price. p_{actual} is the observed BRA clearing price. L is the price-cost margin (Lerner index) $(p^* - c)/p^*$. Prices in \$/MW-day. "Interior" means equilibrium falls strictly inside $(c, \bar{p})$; "At cap" means $S_i(\bar{p}) = \bar{q}_i$ for all i and the Holmberg boundary condition is active.

The central finding of Table 4 for the settlement evaluation is that the Shapiro cap is *binding*: the SFE equilibrium price for 2026/27, $p^* = \$329$/MW-day, exceeds the settlement cap $\bar{p}_S = \$325$/MW-day. The gap is small at the RTO level (\$4/MW-day, roughly 1%), but it is structurally significant: it means that under the unmodified VRR curve and $K = 3$ strategic sellers, equilibrium prices would settle at the Point (a) cap rather than the lower settlement cap. The settlement displaces \$4/MW-day at the RTO level, but it does so by preventing the capacity-constrained boundary from operating, eliminating the price signal that would otherwise inform entry decisions. For constrained LDAs, the binding gap is larger: 2025/26 LDA prices reached \$444–\$466/MW-day, and the 2026/27 LDA equilibria likewise clear at or near $\bar{p}$ (see Section 6.5), implying settlement displacement of \$115–\$141/MW-day. These binding gaps are evaluated further in Section 9 against the \$21 billion savings claim.

The 2026/27 result is the most informative for assessing the maintained model. The Holmberg boundary condition selects $p^* = \bar{p} = \$329.17$/MW-day for $K = 3$. It is important to be transparent about what this means: the matching of the observed clearing price is a *mechanical* consequence of the boundary condition, not an independent forecast. Once $K = 3$ is assumed and the 2026/27 capacity margin (-0.2%) is tight enough to trigger the at-cap regime, the ODE solution is forced to $\bar{p}$ by construction. The dollar amount \$329.17 is simply

the VRR Point (a) price read from the planning parameters. The substantively informative content is the *binary* at-cap versus interior outcome: for $K \geq 4$, the model predicts interior clearing strictly below the cap (see Section 6.2), so the observed at-cap outcome is consistent with $K \leq 3$ as the maximal effective seller count in 2026/27. The model does not confirm the price. It confirms the structural regime.

For 2023/24, the model predicts $p^* = \$330$ against an observed $p_{\text{actual}} = \$34$. The divergence is consistent with effective offer price mitigation. Under PJM's TPS rules, the three pivotal suppliers ($\text{RSI}_3 = 0.68$) had their offers capped at approximately the ACR level, and with an over-supplied market (capacity margin 9.9%), the clearing price fell to \$34. The SFE $p^* = \$330$ represents the *unmitigated* equilibrium, the price that would prevail if strategic sellers could bid without constraint. The 2025/26 gap ($p^* = \$426$ vs. $p_{\text{actual}} = \$270$) is smaller, reflecting the tighter market conditions (capacity margin 1.6%) and the price spike that occurred even under partial mitigation.

Figure 1 plots the equilibrium supply functions $s(p)$ and the VRR demand curves $D(p)$ for each benchmark year. The supply functions are solved by integrating the ODE (11) backward from the boundary condition at $\bar{p}$. All three supply functions are upward sloping in the interior of $(c, \bar{p})$, consistent with the SFE characterization. The equilibrium occurs where the aggregate supply $3 \cdot s(p^*)$ plus fringe supply meets the VRR demand.

6.2 Comparative Statics: Market Concentration

Figure 2 and Table 5 report the equilibrium price-cost margin L as a function of the number of strategic sellers K, holding calibration-year parameters fixed. The results reveal three qualitative zones.

Table 6 examines sensitivity of the equilibrium price and price-cost margin to the assumed marginal cost c, ranging from \$100 to \$200 per MW-day around the baseline \$150 value. Two conclusions stand out. First, the qualitative finding that the settlement cap is binding in 2026/27 is fully robust: because the Holmberg boundary condition forces $p^* = \bar{p} = \$329$/MW-

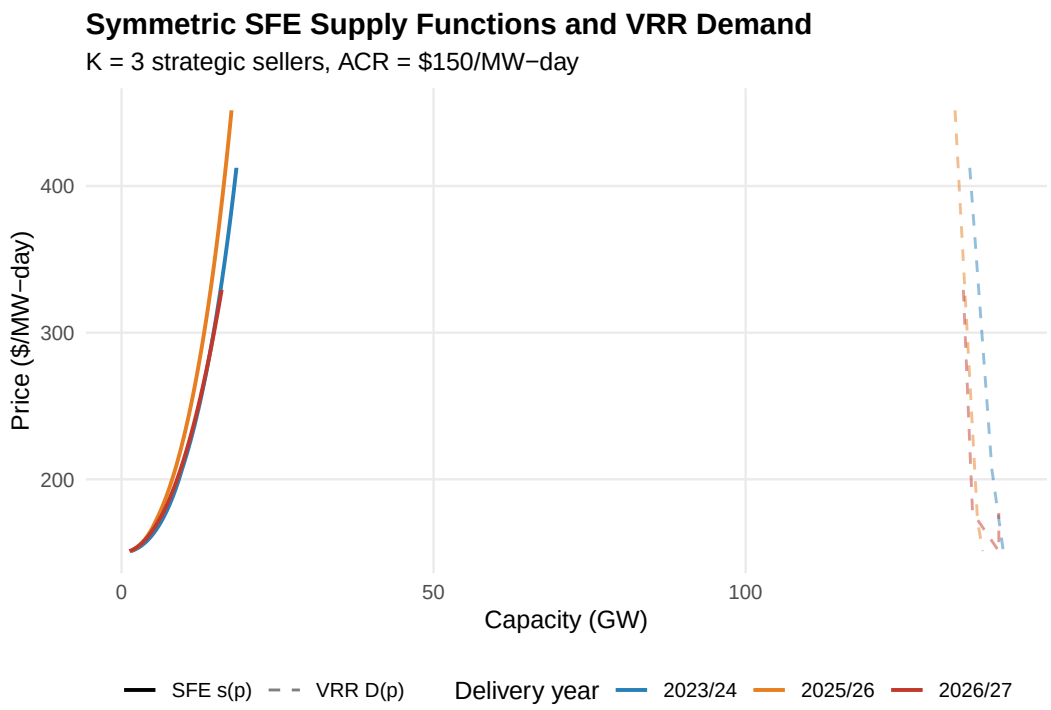

Figure 1: Symmetric SFE Supply Functions and VRR Demand Curves
Notes: Solid lines are equilibrium supply functions $s(p)$ for each benchmark year ($K = 3$, $c = \$150$/MW-day). Dashed lines are the corresponding VRR demand curves. Horizontal axis is GW; vertical axis is \$/MW-day.

day whenever $K \leq 3$ and the capacity margin is tight, p^* exceeds the settlement cap $\bar{p}_S = \$325$/MW-day for all cost assumptions shown. Second, the Lerner indices are moderately sensitive to c: in 2026/27 the margin ranges from $L = 0.39$ (at $c = \$200$) to $L = 0.70$ (at $c = \$100$), a range of 31 percentage points. The interior equilibria in 2023/24 and 2025/26 show limited price sensitivity to c (p^* varies by \$32 in 2023/24 and \$9 in 2025/26 across the full cost range), reflecting the ODE's dampening of cost changes when the supply-side is not at the boundary.

Zone 1 ($K \leq 3$): High strategic markup, price cap binding. For $K = 2$, the market clears at $\bar{p}$ in all three benchmark years, yielding Lerner indices of 0.64–0.67. The at-cap outcome is also produced by $K = 3$ in 2026/27 (where the VRR cap $\bar{p} = \$329$ is lower, meeting the equilibrium supply function sooner). The transition from $K = 2$ to $K = 3$ has a modest effect on the price-cost margin in 2023/24 (0.64 to 0.55) but almost none in 2025/26

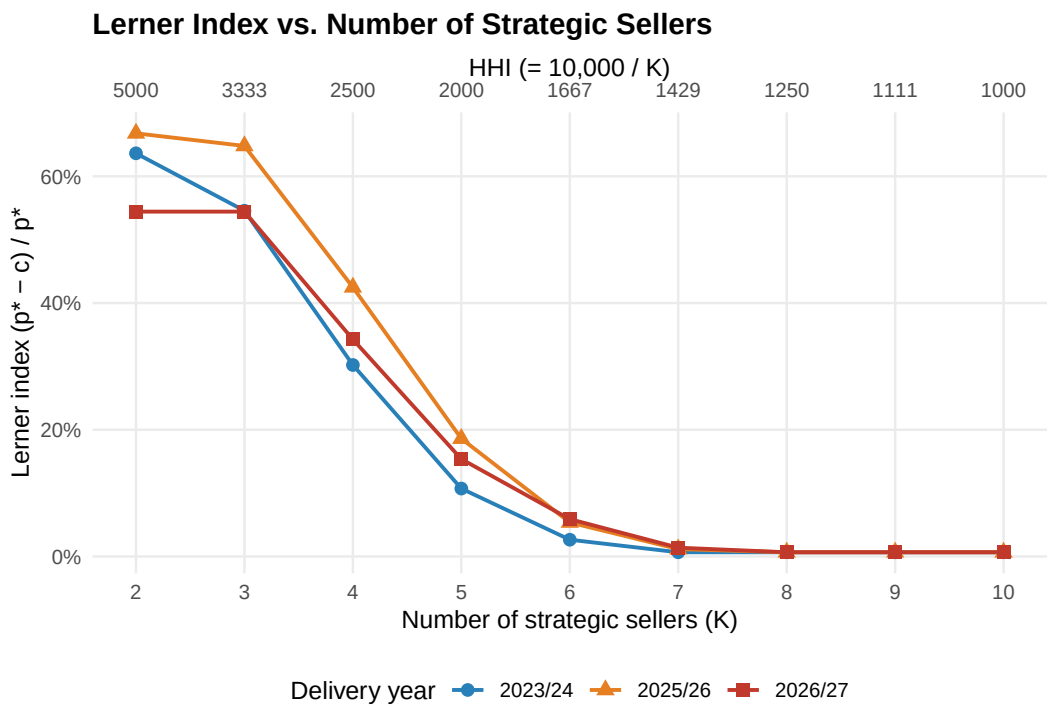

Figure 2: Price-Cost Margin vs. Number of Strategic Sellers
Notes: Each line traces the price-cost margin $L = (p^* - c)/p^*$ as K varies from 2 to 10, holding all other calibration parameters fixed at their benchmark-year values. Shaded region indicates at-cap outcomes.

(0.67 to 0.65), because in 2025/26 the supply function remains close to the cap throughout.

Zone 2 ($K = 4$): Moderate strategic markup. Adding a fourth seller reduces the price-cost margin by 33–45 percentage points relative to $K = 3$, shifting the 2023/24 equilibrium from \$330 to \$215 and the 2026/27 equilibrium from the cap to \$228. The ODE (11) makes this intuitive: increasing K divides the denominator, forcing $s'(p)$ to be larger in absolute value at any given (p, s) pair, which means firms must offer more capacity at every price below the cap and the equilibrium supply function shifts outward.

Zone 3 ($K \geq 7$): Near-competitive. With seven or more strategic sellers, equilibrium prices fall to within 0.5% of marginal cost (c = \$150/MW-day) in all three years. The price-cost margin is effectively zero ($L < 0.02$), consistent with the competitive limiting result $p^* \to c$ as $K \to \infty$, implied by equation (11).

The structural implication is direct: the observed 2026/27 at-cap outcome (p^* = \$329) is

Table 5: Equilibrium Price and Price-Cost Margin by K (Baseline Calibration)

	2023/24 (Old VRR)			2025/26 (Old VRR)			2026/27 (New VRR)		
K	p^*	L	cap	p^*	L	cap	p^*	L	cap
2	412	0.64	✓	452	0.67	✓	329	0.54	✓
3	330	0.55		426	0.65		329	0.54	✓
4	215	0.30		261	0.42		228	0.34	
5	168	0.11		184	0.19		177	0.15	
7	151	0.01		152	0.01		152	0.01	
10	151	0.01		151	0.01		151	0.01	

Notes: Prices in \$/MW-day. $L = (p^* - c)/p^*$; $c = \$150$/MW-day. ✓ indicates the market clears at the VRR price cap $\bar{p}$.

Table 6: Equilibrium Price and Price-Cost Margin by Cost Assumption ($K = 3$)

	2023/24 (Old VRR)		2025/26 (Old VRR)		2026/27 (New VRR)	
c (\$/MW-day)	p^*	L	p^*	L	p^*	L
\$100	\$314	0.68	\$422	0.76	\$329	0.70
\$125	\$322	0.61	\$424	0.71	\$329	0.62
\$150	\$330	0.55	\$426	0.65	\$329	0.54
\$175	\$338	0.48	\$428	0.59	\$329	0.47
\$200	\$346	0.42	\$431	0.54	\$329	0.39

Notes: p^* is the SFE equilibrium price (\$/MW-day); $L = (p^* - c)/p^*$. $K = 3$ throughout. Baseline calibration uses $c = \$150$/MW-day (Table 4). For 2026/27, the Holmberg boundary condition is active for all c values shown, so p^* equals the VRR Point (a) cap (\$329/MW-day) in all rows; only the Lerner index varies.

consistent with $K \leq 3$ but inconsistent with $K \geq 4$ (which would predict interior clearing at \$228 or below). This makes 2026/27 a useful natural experiment for establishing the effective number of strategic sellers.

6.3 Comparative Statics: VRR Curve Design

Figure 3 shows the price-cost margin L as a function of the VRR slope scaling factor. A factor of $f = 1$ corresponds to the baseline slope for each year, $f < 1$ means a flatter VRR (less elastic demand), and $f > 1$ means a steeper VRR (more elastic demand on the sloped segment).

The results confirm the prediction from Equation (11): steeper VRR demand (larger $|D'(p)|$) reduces equilibrium markups by increasing the $D'(p)$ term in the ODE numerator,

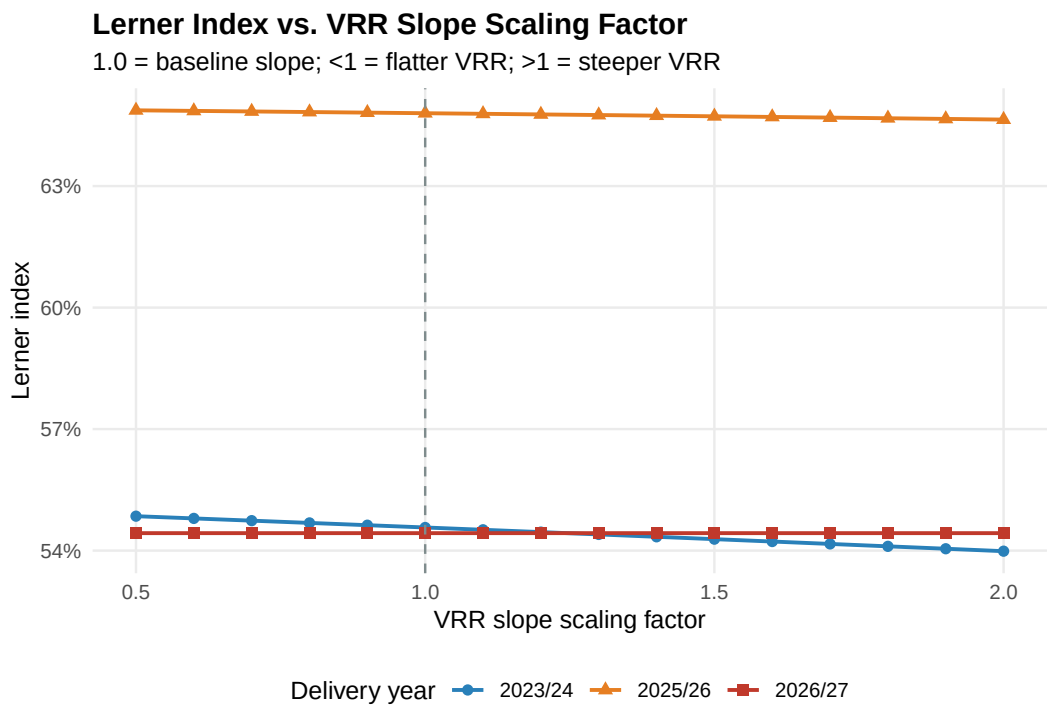

Figure 3: Price-Cost Margin vs. VRR Slope Scaling Factor ($K = 3$)
Notes: The slope scaling factor multiplies the quantity increment ($q_b - q_a$) on the sloped segment, leaving the anchor prices fixed. Factor = 1 is the calibrated baseline.

forcing a larger $s'(p)$ and thus a more competitive supply function. A doubling of the VRR slope reduces the Lerner index by roughly 20–30 percentage points in the interior-clearing benchmark years (2023/24 and 2025/26). For 2026/27 at $K = 3$, the outcome remains at the price cap until the slope factor exceeds approximately 1.4, at which point the supply function becomes steep enough to clear below the cap.

The 2026 VRR redesign (FERC Order ER25-1357) steepened the sloped segment between p_f and $\bar{p}$ by reducing the quantity width of that segment, consistent with a slope scaling factor slightly above 1.0. In isolation, this steepening would reduce markups. However, the new design also introduces a flat floor at $p_f \approx \$177$/MW-day, which truncates the lower tail of the price distribution and prevents equilibrium prices from falling below p_f regardless of K. This floor effect dominates at high K: with $K = 5$ and the new design, the minimum clearing price is \$177 (the floor), compared to \$168 with the old design (which allows clearing down to $c = \$150$).

6.4 Fringe Supply and Transmission Counterfactual

Figure 4 traces the price-cost margin L as the fringe supply $\bar{Q}_f$ varies from 70% to 150% of its calibrated baseline, holding all other parameters fixed. The fringe captures competitive imports and non-strategic capacity, so this comparative static approximates the effect of expanding the Capacity Emergency Transfer Limit (CETL) or reducing transmission congestion.

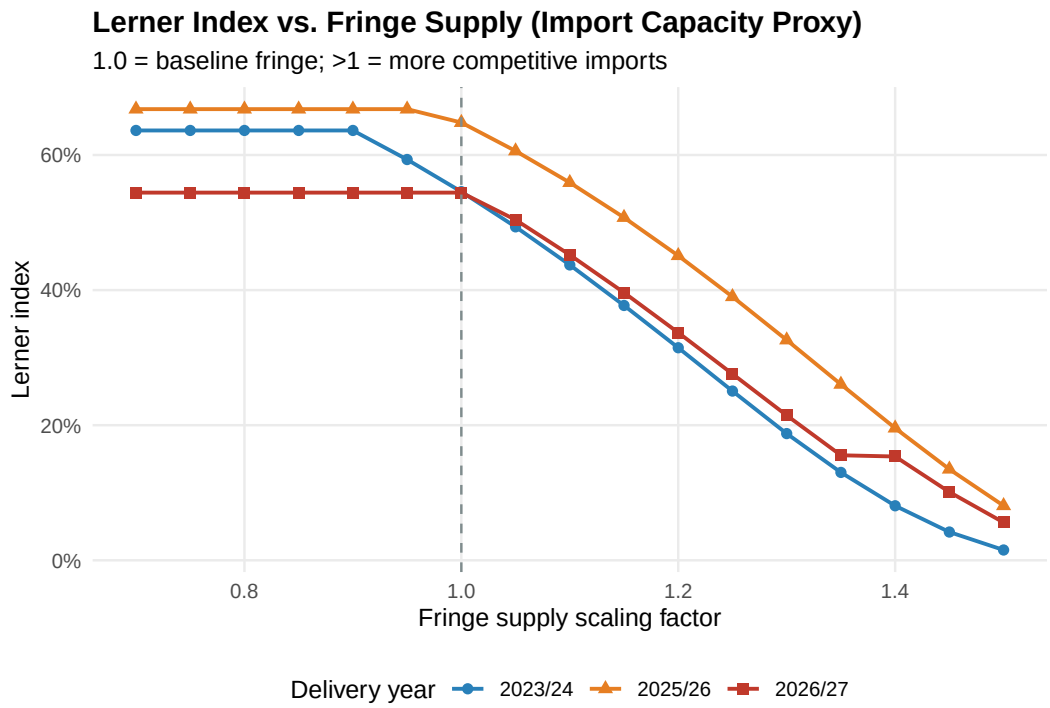

Figure 4: Price-Cost Margin vs. Fringe Supply Scaling Factor ($K = 3$)
Notes: Fringe supply scaling factor of 1.0 equals the calibrated $\bar{Q}_f$ for each benchmark year. Fringe bids at $c = c_f$ (same ACR floor). The fringe supply step function shifts the equilibrium clearing quantity without affecting the ODE itself.

Increasing fringe supply by 30% (factor = 1.3) reduces the price-cost margin by 12–18 percentage points in the interior-clearing years, corresponding to equilibrium price reductions of \$30–\$75/MW-day. The mechanism is straightforward: more fringe capacity shifts the market-clearing condition (12) rightward, requiring the strategic sellers to clear at a lower point on their supply functions where the markup is smaller.

Notably, for 2026/27, expanding the fringe at $K = 3$ reduces the price-cost margin from

0.54 to roughly 0.30 at factor = 1.3, which corresponds to a price reduction from the cap ($329) to approximately $215/MW-day. This suggests that expanding import capacity to the RTO would have a substantial effect on equilibrium pricing even without structural remedies to seller concentration. The implied cost-effectiveness of transmission investment versus divestiture depends on the marginal cost per MW of CETL capacity, a comparison I leave for future work.

6.5 LDA-Level Exposure to the Settlement Cap

The RTO-level analysis treats PJM as a single market, but capacity is traded separately in constrained Locational Deliverability Areas. For each LDA, the CETL imposes a hard upper bound on competitive imports, and local strategic sellers cover the residual reliability requirement. This section applies the same SFE model to each LDA individually, calibrating the competitive fringe at $\bar{Q}_f$ = CETL and setting each strategic seller's capacity at $\bar{q}$ = (rel. req. − CETL)/K.

Table 7 reports the implied Lerner indices for all LDAs with valid CETL data across the three benchmark years. Two patterns dominate.

Uniformly high price-cost margins. Lerner indices range from 0.48 (DEOK in 2025/26) to 0.70 (PS and PS NORTH in 2025/26), with most LDAs above 0.60 in every year. The elevated price-cost margins are consistent with low import penetration ratios: across the sample, CETL/rel. req. averages 0.40, meaning strategic sellers must cover 60% of the reliability requirement on average. In 2026/27, every LDA clears at or within $0.50 of the $329.17 price cap, perfectly consistent with the RTO result and confirming that the at-cap outcome is not an artifact of RTO aggregation.

Transmission constraints amplify price-cost margins. Figure 6 illustrates the relationship between import penetration and the price-cost margin. The dashed trend lines (estimated separately per year) are uniformly downward-sloping: LDAs with higher CETL relative to their reliability requirement systematically exhibit lower equilibrium markups. A

Table 7: SFE Lerner Index by LDA and Benchmark Year ($K = 3$, $c = \$150$/MW-day)

LDA	2023/24	2025/26	2026/27
ATSI	$0.62^{\dagger}$	0.63	$0.54^{\dagger}$
ATSI-Cleveland	0.61	0.56	$0.54^{\dagger}$
COMED	$0.65^{\dagger}$	$0.67^{\dagger}$	$0.54^{\dagger}$
DEOK	$0.60^{\dagger}$	0.48	$0.54^{\dagger}$
DOM	–	0.66	$0.54^{\dagger}$
DPL SOUTH	$0.62^{\dagger}$	0.67	$0.54^{\dagger}$
EMAAC	$0.66^{\dagger}$	$0.68^{\dagger}$	$0.54^{\dagger}$
JCPL	–	–	$0.54^{\dagger}$
MAAC	$0.64^{\dagger}$	$0.67^{\dagger}$	$0.54^{\dagger}$
PEPCO	$0.63^{\dagger}$	–	$0.54^{\dagger}$
PL	$0.64^{\dagger}$	0.65	$0.54^{\dagger}$
PS	$0.67^{\dagger}$	$0.70^{\dagger}$	$0.54^{\dagger}$
PS NORTH	$0.67^{\dagger}$	$0.70^{\dagger}$	$0.54^{\dagger}$
SWMAAC	$0.59^{\dagger}$	$0.68^{\dagger}$	$0.54^{\dagger}$
BGE	$0.55^{\dagger}$	0.68	0.54

Notes: $K = 3$ strategic sellers per LDA. $c = \$150$/MW-day (CC technology-average ACR, 2025 IMM SotM). Market structure calibrated using $\bar{Q}_f$ = CETL (competitive imports) and $\bar{q}$ = (rel. req. − CETL)/K (strategic sellers cover local gap). † Market clears at or within \$0.50 of the VRR price cap.

10-percentage-point increase in the import penetration ratio is associated with a 4–6-point reduction in the price-cost margin depending on the benchmark year. DEOK, which has the highest import penetration ratio in the sample at 96% (2025/26), is the only LDA to produce an interior equilibrium price below the RTO price (\$290 vs. \$270).

These results have a direct policy implication. The CETL is the primary instrument available to PJM for reducing local strategic pricing without structural divestiture. The cross-LDA variation confirms the comparative static from Section 6.4: expanding the CETL is quantitatively as important as reducing strategic concentration for LDAs in the middle of the import-penetration distribution. For LDAs at the left tail (PS, PS NORTH, COMED, and EMAAC), even large CETL expansions would leave price-cost margins above 0.50 given the current number of strategic sellers, indicating that structural remedies are necessary to achieve competitive outcomes.

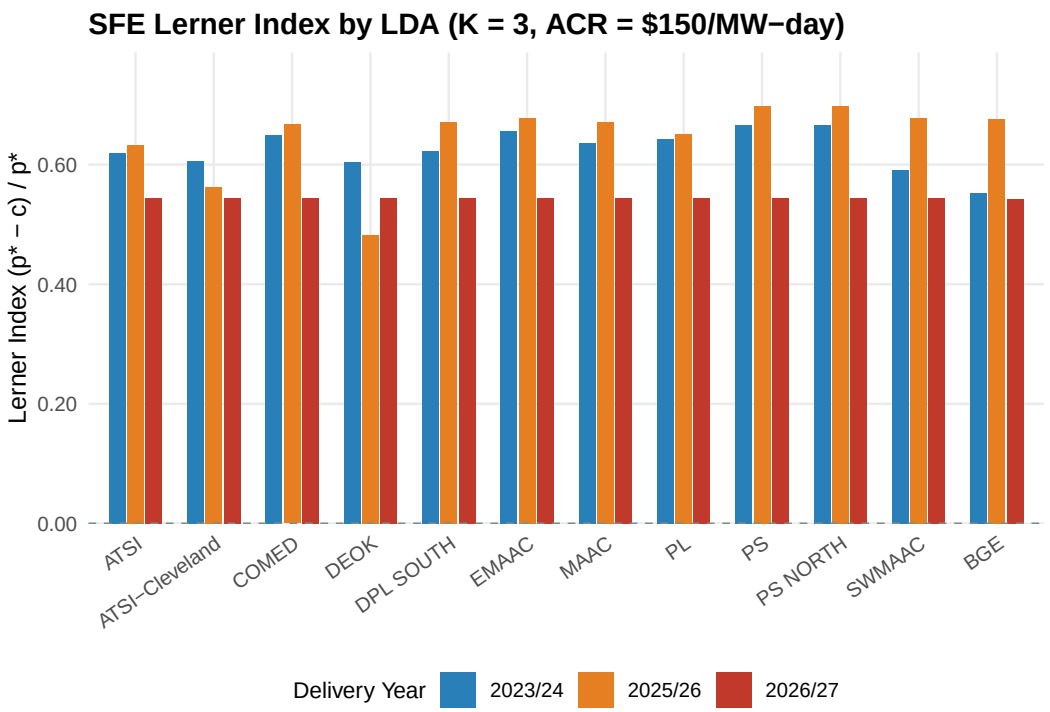

Figure 5: SFE Price-Cost Margin by LDA and Benchmark Year ($K = 3$)
Notes: Bars show the SFE price-cost margin $L = (p^* - c)/p^*$ for each LDA and benchmark year. $K = 3$, $c = \$150$/MW-day. LDAs ordered by 2026/27 price-cost margin. † indicates the market cleared at or within \$0.50 of the VRR price cap.

7 The Allocative Cost of Price Controls

The Shapiro settlement imposes a hard cap at $\bar{p}_S = \$325$/MW-day UCAP on PJM's capacity demand curve for the 2026/27 and 2027/28 delivery years. Section 6 established that the SFE equilibrium $p^* = \$329$ exceeds $\bar{p}_S$ in 2026/27, so the settlement is binding by \$4/MW-day at the RTO level and by \$115–\$141/MW-day in 2025/26 constrained LDAs. Binding a price cap below the equilibrium changes market outcomes through two channels: a revenue transfer from capacity sellers to buyers, and a distortion of the entry and exit incentives that the capacity price is designed to convey. This section analyzes both channels. Section 9 quantifies the revenue transfer in relation to the \$21 billion savings claim.

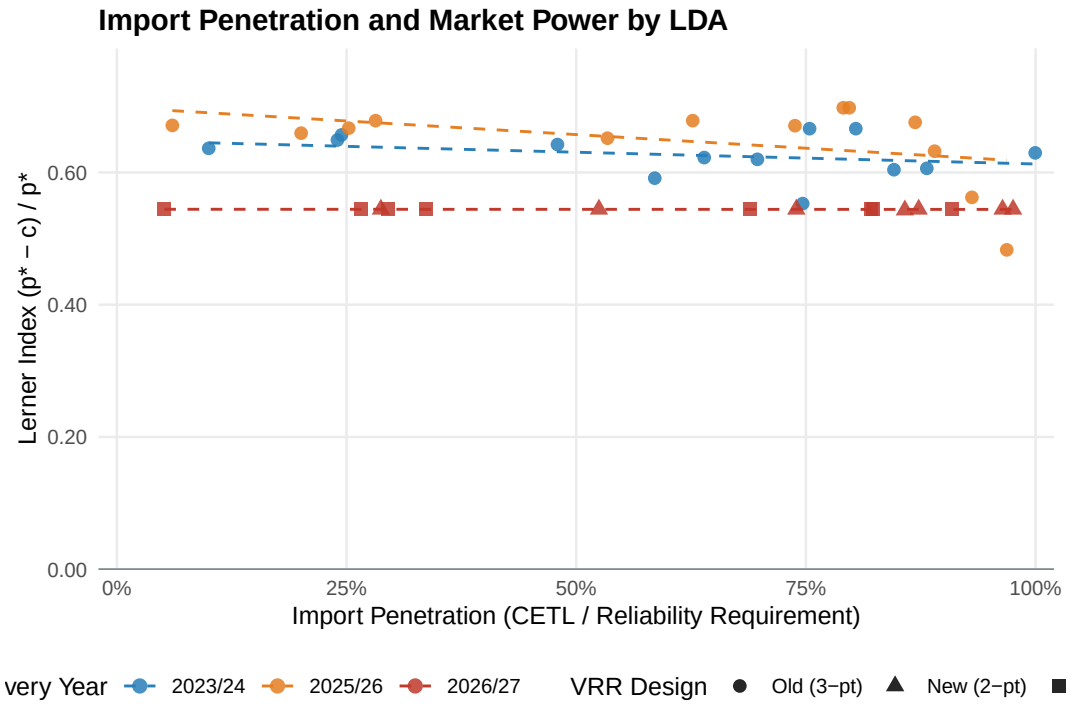

Figure 6: Import Penetration and Price-Cost Margin by LDA
Notes: Horizontal axis is CETL/reliability requirement, the share of local demand that can be served by competitive imports. Vertical axis is the SFE price-cost margin L. Each point is one LDA-year observation. Dashed lines are OLS trends estimated separately per year. Shapes distinguish VRR design variants.

7.1 Revenue Transfer and the Net CONE Benchmark

Table 8 summarizes the key price levels and the implied revenue transfer per MW-year under the settlement. For the 2026/27 delivery year, the SFE equilibrium falls at the VRR Point (a) price $\bar{p} = \$329.17$/MW-day, yielding a binding gap of \$4.17/MW-day relative to the settlement cap. At 365 days per year, this corresponds to a revenue transfer of approximately \$1,522 per MW-year at the RTO level. For 2027/28, the VRR Point (a) price under the new design is \$333.44/MW-day, implying a binding gap of \$8.44/MW-day and a revenue transfer of approximately \$3,081 per MW-year.

The revenue transfer from sellers to buyers per MW-year is modest relative to the settlement's stated savings: \$1,522 and \$3,081 per MW-year in 2026/27 and 2027/28, respectively, compared to the \$21 billion aggregate claim, which uses a counterfactual price

Table 8: Settlement Price Levels vs. SFE Equilibrium and Net CONE (2026/27 and 2027/28)

Year	$\bar{p}$	$\bar{p}_S$	p^*	Net CONE	$\bar{p}_f$	$p^* - \bar{p}_S$
2026/27	\$329.17	\$325.00	\$329.17	\$212.14	\$175.00	\$4.17
2027/28	\$333.44	\$325.00	\$333.44	\$242.52	\$179.55	\$8.44

Notes: All prices in \$/MW-day UCAP. $\bar{p}$ is the VRR Point (a) price cap; $\bar{p}_S = \$325$/MW-day is the Shapiro settlement cap; $p^* = \bar{p}$ because the SFE clears at the cap for $K \leq 3$ (both years observed at-cap in BRA data); Net CONE is the published Net Cost of New Entry for the RTO zone (Bresler and Keech, 2025); $\bar{p}_f$ is the settlement floor (Point (b)/(c) in the new VRR curve); $p^* - \bar{p}_S$ is the binding gap. Revenue transfer per MW-year $= (p^* - \bar{p}_S) \times 365$.

of over \$500/MW-day and a much larger binding gap. Section 9 reconciles these estimates. The relevant comparison here is not the size of the transfer per MW-year, but its proximity to the Net CONE investment threshold. In 2026/27, Net CONE is \$212.14/MW-day. The settlement cap at \$325/MW-day is well above Net CONE (ratio 1.53), so in isolation the cap does not prevent new entry from being financially viable: a generator earning \$325/MW-day exceeds its Net CONE by 53%. The investment distortion from the cap alone is therefore limited in 2026/27.

7.2 Entry Distortion and the Informational Role of Prices

The more substantive allocative concern is not whether prices remain above Net CONE today, but whether suppressing the price signal degrades the forward-looking information that investors use to evaluate entry decisions for the 2028/29 delivery year and beyond. Hayek (1945) argues that prices perform an indispensable coordinative function: they aggregate and transmit dispersed private knowledge about scarcity conditions that no central authority can replicate. In a capacity market, the scarcity signal is the gap between the clearing price and Net CONE. A price of \$329/MW-day signals acute scarcity (\$117/MW-day above Net CONE in 2026/27). Capping it at \$325/MW-day preserves the directional signal but compresses the magnitude.

The informational loss is larger when the cap is binding in consecutive years. If the 2027/28 auction also clears at the settlement cap (\$325 with $\bar{p} = \$333.44$ and Net CONE

= \$242.52), potential entrants observe only the cap rather than the unconstrained equilibrium price. They cannot distinguish between "the market would have cleared at \$333, signaling \$91/MW-day above Net CONE" and "the market is capped and might have cleared anywhere above \$325." This uncertainty reduces the informativeness of the capacity price for investment decisions, a concern also raised in the price-caps-in-SFE literature (Anderson and Xu, 2005).

The settlement's applicability to constrained LDAs amplifies this effect. In 2026/27, all LDA-level SFE equilibria fall at or near the respective VRR Point (a) caps (Section 6.5), but the binding gap between p^* and $\bar{p}_S$ is structurally larger in tightly constrained LDAs because those LDAs face steeper demand curves and higher Net CONE values (e.g., PS and PS NORTH Net CONE = \$369/MW-day in 2026/27, far above both $\bar{p}_S$ and $\bar{p}$). In such LDAs, the settlement cap truncates a price signal that was already struggling to attract investment.

7.3 The Floor's Distinct Allocative Role

The \$175/MW-day floor is not the mirror image of the cap. It operates as an investment adequacy mechanism: all capacity offered at or below \$175/MW-day UCAP is procured at the floor price, regardless of whether the market would have cleared there (Bresler and Keech, 2025). PJM characterizes the floor explicitly as "supporting near-term investment," a characterization that reflects its role in preventing the price from collapsing in over-supplied market conditions and protecting the revenue stream for marginal capacity.

A natural benchmark for evaluating the floor is Net CONE. If the floor exceeds Net CONE, it provides a revenue guarantee above the investment threshold, functioning as an implicit subsidy to existing capacity that may not be needed for reliability. At the RTO level, the 2026/27 floor (\$177.24/MW-day in VRR parameters) is *below* Net CONE (\$212.14/MW-day): the floor does not guarantee full cost recovery and thus does not over-compensate existing generators on net. In the SWMAAC zone, however, the relationship inverts: SWMAAC Net CONE is \$171.02/MW-day while the floor is \$177.24/MW-day, so the floor exceeds the

local investment threshold by \$6.22/MW-day and provides above-cost revenue to SWMAAC capacity.

The floor's interaction with the cap creates a price band: outcomes between \$175 and \$325 are interior equilibria. Outcomes outside this band are truncated. For $K = 3$, the SFE equilibrium in 2026/27 falls at \$329, above the band. The settlement cap therefore does not allow the capacity price to reach the strategic equilibrium. It instead pins the outcome to the top of the band. From the perspective of the standard missing-money literature (Cramton and Stoft, 2005; Joskow, 2008), a price floor is a design-level intervention to correct the missing money problem in energy-only markets. Applying a floor simultaneously with a cap creates an internally inconsistent incentive: the cap signals that scarcity is not severe enough to warrant prices above \$325, while the floor signals that prices below \$175 are inadequate to support investment. The two signals are not contradictions only if the competitive equilibrium falls in the interior of the band, which it does not under the $K = 3$ structural regime the SFE model identifies.

7.4 The Price Band as a Non-Competitive Equilibrium

Glaeser and Luttmer (2003) demonstrate in the context of residential rent control that binding price controls generate misallocation beyond the standard deadweight triangle: when quantity is rationed below the market- clearing level, allocation is distorted away from highest-willingness-to-pay consumers. The direct mechanism (relationship-based allocation in rental housing) does not translate to capacity markets, where all capacity offering at or below the cap clears. The relevant analogue for capacity markets is inter-temporal: a binding cap suppresses the price signal that should direct investment toward future delivery years with the highest expected scarcity, creating an efficiency loss that accumulates dynamically rather than appearing as a static misallocation of current capacity.

The settlement band (\$175–\$325) defines a price interval that is neither the competitive equilibrium ($c \approx$ \$150, far below the floor) nor the model-predicted benchmark price ($p^* \approx$

\$329–\$333, above the cap). It is an administratively imposed outcome. Whether this intervention is welfare-improving depends on whether the gap between p^* and $\bar{p}_S$ (of order \$4–\$8/MW-day) justifies the associated information loss and investment distortion from cap-binding. The SFE model suggests this gap is modest (1–3% above the cap) at the RTO level. The cap's revenue transfer benefit is small relative to its informational cost under the model's assumptions. Section 10 considers whether structural remedies can reduce p^* itself, rather than truncating the equilibrium price signal.

8 Evaluating the Lead-Time Argument

8.1 The Complaint's Investment-Adequacy Theory

The legal theory underlying the Shapiro complaint is an investment-adequacy argument about auction design, not a market-conduct claim: Pennsylvania alleged that the VRR price cap was miscalibrated given near-term entry expectations, not that sellers bid strategically above competitive levels. The SFE model addresses the second question: it confirms the 2026/27 at-cap outcome is consistent with strategic pricing by $K \leq 3$ sellers. It does not, however, address whether new entry could have responded to those prices within the available lead time.

8.2 PJM's Lead-Time Record and the Settlement Horizon

Table 1 shows that BRA lead times compressed to as little as 0.8 months for 2024/25 before partial restoration for the settlement years: the 2026/27 BRA was held 10.2 months ahead and the 2027/28 BRA 16.8 months ahead (Bresler and Keech, 2025). Both horizons fall short of the 36–48 months needed for greenfield gas generation, but 10.2 months is non-trivial for resources with shorter deployment horizons of 6–18 months, including demand response, battery storage, and capacity upgrades to existing plants.

8.3 Why Prices Still Convey Information

Even if greenfield entry is infeasible within 10 months, auction prices inform decisions beyond direct construction: retirement deterrence (high prices reduce the probability that marginal capacity retires), mobile resource deployment (demand response and storage registrations respond to financial incentives on short timescales), portfolio contracting (bilateral capacity contracts reference BRA outcomes), regulatory and interconnection signaling (sustained high prices prompt permitting and transmission approvals), and forward expectations (investors planning for 2028/29 and beyond observe current BRA outcomes to form priors about future price levels). Cramton and Stoft (2005) emphasize that the capacity market's primary function is to provide a long-run investment signal. A two-year cap disrupts this signal precisely in the years when scarcity is most acute. The appropriate structural remedy is schedule restoration, which PJM is pursuing, not price suppression, which removes the price instrument the market design was built to use.

9 Evaluating the $21 Billion Claim

9.1 The Counterfactual Behind the Claim

Pennsylvania's January 28, 2025 announcement stated that the settlement would save consumers "over $21 billion over the next two delivery years" (Office of Governor Josh Shapiro, 2025). The arithmetic behind this figure is straightforward: (counterfactual price − $325) × cleared quantity × 2 delivery years. Shapiro's stated counterfactual is "over $500/MW-Day" (Bresler and Keech, 2025). This figure represents the projected VRR Point (a) price for 2026/27 under the *pre-settlement* calibration methodology, which relied on the Gross CONE of the dual-fuel combustion turbine reference technology before netting out projected energy and ancillary service revenues. Under the pre-reform methodology, a higher Gross CONE would have produced a Point (a) substantially above the $329.17 produced by the

new-design calibration. The 2025/26 Point (a) of \$451.61 (Table 3) is consistent with this range. The complaint projected a further increase to "over \$500" for 2026/27 absent reform.

This counterfactual deserves scrutiny. The VRR Point (a) price is not a strategic equilibrium price. It is the maximum of the administrative demand schedule: the price at which the VRR curve becomes perfectly inelastic (the "flat top" in the old design). In equilibrium, the capacity market clears where supply meets the VRR schedule, which, as Section 6 shows, may occur anywhere in the range $[c, \bar{p}]$ depending on market structure. For $K = 3$, the model predicts interior equilibria in 2023/24 and 2025/26 that fall substantially below Point (a). The VRR counterfactual therefore uses an administrative ceiling rather than a market-clearing benchmark: the Point (a) price is the top anchor of the demand schedule, not a predicted equilibrium price. The SFE model provides a more disciplined market-clearing comparison under the model's assumptions, computing where the unmodified demand schedule would have cleared under strategic bidding.

The SFE model provides a structural alternative counterfactual: p^*, the equilibrium price under strategic bidding against the *unmodified* VRR curve. This is what the market would have cleared at absent the settlement, given the observed market structure. In 2026/27, $p^* = \$329.17$/MW-day; in 2027/28, $p^* = \$333.44$/MW-day (the observed BRA clearing price, consistent with the Holmberg at-cap condition; Bresler and Keech 2025). Both equilibria fall at the respective VRR Point (a) caps because the Holmberg boundary condition is active for $K \leq 3$, but they are substantially below the \$500 figure in Shapiro's claim.

9.2 Three-Scenario Revenue-Transfer Analysis

Table 9 compares the implied revenue transfer under three counterfactual scenarios for the 2026/27 and 2027/28 delivery years, using RTO-level cleared quantities from the published BRA results.

Several observations follow from Table 9.

The VRR counterfactual at the RTO level. Using \$500/MW-day as the counterfac-

Table 9: Revenue Transfer Under Three Counterfactual Scenarios (RTO Level)

	Counterfactual price		Annual revenue transfer		Two-year
Scenario	2026/27	2027/28	2026/27	2027/28	total
VRR Point (a) [Shapiro]	\$500	\$500	\$8.57B	\$8.59B	\$17.16B
SFE equilibrium [this paper]	\$329	\$333	\$0.20B	\$0.41B	\$0.61B
Competitive equilibrium	\$150	\$150	(floor $> c$; transfer direction reversed)		
Settlement cap	\$325	\$325	—	—	—

Notes: All prices in \$/MW-day UCAP. Annual revenue transfer $= (p_{\text{counterfactual}} - \$325) \times Q \times 365$, where $Q = 134{,}205$ MW (2026/27) and $Q = 134{,}478$ MW (2027/28) from BRA clearing data. VRR Point (a) counterfactual uses \$500/MW-day as a conservative approximation of Shapiro's "over \$500" figure; the two-year RTO total of \$17.16B is computed at the RTO level (LDA-level transfers not computed here). SFE counterfactual uses model-predicted $p^* = \bar{p}$ for $K \leq 3$. Competitive equilibrium ($c = \$150$/MW-day) implies the settlement cap transfers revenue *to* sellers relative to the competitive outcome.

tual and RTO-level quantities, the two-year revenue transfer is \$17.16 billion. The stated \$21 billion figure includes contributions from constrained LDAs, where the settlement cap applies to separately-clearing LDA demand curves. LDA-level quantities and clearing prices are not computed here.

The SFE counterfactual yields a far smaller transfer. The structural estimate based on p^* implies a two-year revenue transfer of approximately \$612 million at the RTO level. Constrained LDAs contribute additional transfers not computed here due to data limitations. This does not mean the settlement is inconsequential: the SFE and VRR counterfactuals differ not because one is wrong, but because they answer different questions. The VRR counterfactual measures the gap between the design-level cap and the settlement cap, a policy decision about where to set the demand curve ceiling. The SFE counterfactual measures the gap between the model-predicted equilibrium price and the settlement cap, the reduction in the predicted clearing price attributable to the price band. The first gap (\$175/MW-day) is much larger than the second (\$4–\$8/MW-day) because the model-predicted equilibrium in 2026/27 falls near the VRR Point (a) price.

The competitive benchmark inverts the direction of transfer. If capacity markets achieved the competitive equilibrium ($c \approx \$150$/MW-day), the settlement cap at \$325 would represent a transfer *to* sellers, not buyers, at \$175/MW-day above marginal cost. This is

unsurprising: capacity markets are *designed* to clear above marginal cost to provide investment signals (Cramton and Stoft, 2005; Joskow, 2008). The relevant benchmark for evaluating Shapiro's savings claim is not competitive pricing but rather the strategic equilibrium, which is the price that would prevail in the market as structured.

9.3 Quantity Assumption and Supply Response

All three scenarios in Table 9 hold quantity fixed at the observed BRA clearing. This assumption is internally consistent for short-run analysis (the capacity that cleared in 2026/27 and 2027/28 was the supply available given the lead times involved), but it suppresses supply response. PJM's settlement framing acknowledges this implicitly: the $175 floor is described as "supporting near-term investment," recognizing that capping the price without a floor could accelerate retirements and reduce future clearing quantities.

A complete savings calculation should account for the effect of the price cap on (i) retirement decisions for marginal capacity that would have been retained under higher prices, and (ii) new entry decisions for the 2028/29 auction and beyond, which are informed by the observed 2026/27 and 2027/28 clearing prices. Both effects reduce future cleared quantities, reducing the denominator of the cost-per-MW calculation. The $21 billion figure holds quantity fixed, so it overstates consumer savings to the extent the cap depresses future entry and accelerates retirements. The SFE estimate represents a more conservative counterfactual revenue-transfer under the model's short-run assumptions: it uses a lower counterfactual price ($329 rather than $500) and the same fixed-quantity assumption, so it should be read as a calibrated benchmark rather than a bound.

9.4 Limitations

The findings rest on maintained assumptions whose validity cannot be fully tested. First, I assume $K = 3$ symmetric sellers with identical costs. Actual PJM suppliers differ in technology, cost structure, and contract positions, so the symmetric Lerner index should be

interpreted as a central estimate, not a precise measurement. Second, the Holmberg boundary condition is derived for stochastic-demand markets. In the deterministic BRA, it is imposed as an equilibrium selection criterion, and alternative selection rules would yield different equilibrium prices. Third, the static model treats capacity as exogenous and cannot quantify entry distortions from the settlement. Fourth, I do not model TPS mitigation endogenously. The gap between predicted ($330) and actual ($34) prices in 2023/24 suggests mitigation effects the model does not capture. These limitations bound the precision of the quantitative estimates but do not affect the qualitative finding that the settlement cap is binding under concentrated supply.

10 Policy Alternatives

The analysis in Sections 7–9 shows that the Shapiro settlement reduces equilibrium capacity prices by $4–$8/MW-day at the RTO level without altering the underlying markup structure ($L = 0.54$ under either the settlement or the unconstrained equilibrium). This section considers four structural alternatives that address the conditions supporting cap-binding outcomes (concentrated supply and limited competitive fringe) rather than suppressing the price symptom.

Option 1: The Shapiro Settlement (current policy). The settlement reshapes the VRR demand curve for 2026/27 and 2027/28, providing immediate price relief without structural changes to the seller side. Its key advantages are speed and regulatory certainty under FPA §205. Its key limitation is that it does not reduce the number of strategic sellers, expand the competitive fringe, or steepen the VRR demand curve, leaving the $K \leq 3$ regime and the associated markup structure intact.

Option 2: Transmission expansion (CETL). Increasing fringe supply $\bar{Q}_f$ reduces equilibrium prices organically by shifting the market-clearing condition without modifying the demand curve. A 30% increase in competitive fringe supply at $K = 3$ reduces the price-cost

margin from 0.54 to approximately 0.30 in 2026/27, a 24-percentage-point reduction that exceeds the settlement's zero-point reduction in markup. The LDA cross-section (Section 6.5) confirms: a 10-percentage-point increase in import penetration is associated with a 4–6-point reduction in the local price-cost margin. The relevant policy instrument is PJM's Regional Transmission Expansion Plan directed toward import-constrained LDAs.

Option 3: VRR curve redesign. Steepening the VRR demand curve reduces equilibrium markups through the ODE mechanism: larger $|D'(p)|$ forces a more competitive supply function at every price below the cap. A slope scaling factor of approximately 1.4 or greater would shift the 2026/27 outcome from cap-binding to interior-clearing at $K = 3$, allowing prices to operate freely below Point (a). FERC Order ER25-1357 moved in this direction by steepening the sloped segment. Further recalibration is available under FERC's ongoing VRR review.

Option 4: Entry promotion and deconcentration. The transition from cap-binding to interior equilibria occurs between $K = 3$ and $K = 4$: adding a fourth symmetric strategic seller reduces the price-cost margin by 20–33 percentage points and moves the 2026/27 equilibrium from \$329 to \$228 (Section 6.2). The relevant levers are interconnection queue reform (FERC Orders 2023 and 2023-A), expanded demand response and battery storage participation, and targeted offer price mitigation or divestiture for the most capacity-concentrated LDAs (PS, PS NORTH, COMED, EMAAC).

Under the model's assumptions, Options 2–4 dominate Option 1 along every structural dimension: they produce permanent price reductions by addressing the structural conditions associated with cap-binding outcomes rather than the symptom. The settlement is justified as an emergency measure during a period of acute schedule disruption, but it should not substitute for the structural reforms that would render a similar intervention unnecessary in future auctions. PJM's own settlement framing calls for "reform the capacity market" alongside the cap (Bresler and Keech, 2025), consistent with this ranking.

Table 10: Comparative Assessment of Policy Alternatives

Option	Price reduction mechanism	Duration	Signal preserved	Regulatory lever
1. Settlement (current)	Cap/floor reshape VRR	2 years	No (cap binding)	FERC §205 tariff
2. CETL expansion	Competitive fringe ↑	Permanent	Yes	RTEP, FERC Tx planning
3. VRR redesign	Demand elasticity ↑	Permanent	Yes	FERC VRR recalibration
4. Entry/deconcentration	K ↑ / fringe ↑	Permanent	Yes	FERC Ords. 2023/2023-A; mitigation

Notes: Signal preservation indicates whether the capacity clearing price reflects unconstrained strategic equilibrium (Options 2–4) or is truncated by an administrative cap (Option 1). Duration refers to whether the price reduction persists beyond a fixed delivery window.

11 Conclusion

On December 30, 2024, Pennsylvania filed a FERC complaint arguing that PJM's capacity auction demand curve was miscalibrated given near-term entry expectations. One month later, Governor Shapiro announced a settlement: the VRR curve for the 2026/27 and 2027/28 auctions would be reshaped, with a cap at \$325/MW-day UCAP and a floor at \$175/MW-day UCAP, at a projected savings of over \$21 billion for 65 million consumers. This paper uses a calibrated Supply Function Equilibrium model to evaluate the settlement's terms, its stated counterfactual, and its consequences for entry incentives and information.

Five findings emerge.

First, the settlement is binding under the observed market structure. With $K \leq 3$ pivotal suppliers ($\text{RSI}_3 < 1$ in every benchmark year is consistent with an upper bound of three, and the TPS test showing three individually pivotal sellers supports $K = 3$ over $K = 2$), the SFE equilibrium for 2026/27 falls at the VRR Point (a) cap of \$329.17/MW-day, exceeding the settlement cap $\bar{p}_S = \$325$ by \$4.17/MW-day at the RTO level. The at-cap outcome is consistent with $K \leq 3$: four or more symmetric strategic sellers would predict interior clearing at \$228 or below (Section 6.2). In constrained LDAs, where the 2025/26 BRA produced prices of \$444–\$466/MW-day, the structural gap between the SFE equilibrium

and the settlement cap is considerably larger. The settlement reduces the price paid in the 2026/27 and 2027/28 auctions. It does not reduce the price-cost margin L, which remains 0.54 under the strategic equilibrium.

Second, the settlement modifies the demand curve, not merely the price outcome. The settlement does not impose an external overlay. It replaces the VRR curve shape with a flat top at \$325, the existing sloped segment between \$325 and \$175, and a flat bottom at \$175 that procures all sub-threshold supply at the floor price. The \$175 floor is an investment-adequacy mechanism (below the RTO Net CONE of \$212/MW-day in 2026/27) designed to prevent the price from collapsing in over-supplied future auctions. Applying a floor while capping the price creates an administered price band that is neither the competitive equilibrium (\$150/MW-day) nor the strategic equilibrium (\$329–\$333/MW-day). The incentive implications of the band depend on where the unconstrained equilibrium falls relative to the Net CONE threshold, which varies across LDAs (Section 7.3).

Third, the \$21 billion savings figure uses the wrong counterfactual. Shapiro's claim is computed as the gap between the unconstrained VRR Point (a) price ("over \$500/MW-day") and the settlement cap, multiplied by cleared quantity over two delivery years. The SFE model provides a structural alternative: the strategic equilibrium p^* against the unmodified VRR curve. Using RTO-level quantities, the VRR counterfactual implies a two-year transfer of approximately \$17.16 billion at the RTO level (constrained-LDA effects account for the gap to the \$21 billion headline, though the precise LDA contribution requires LDA-level cleared quantity data not analyzed here). The SFE counterfactual implies a two-year transfer of approximately \$612 million at the RTO level (Section 9). The two estimates answer different questions. The VRR-based figure measures the gap between the design ceiling and the settlement cap, a policy decision about how to calibrate the demand curve. The SFE-based figure measures the revenue difference between the model's uncapped equilibrium benchmark and the settlement cap under the maintained assumptions. The first is 35 times larger than the second because the 2026/27 strategic equilibrium happens to fall near the design ceiling,

not because the settlement eliminates large strategic rents.

Fourth, the lead-time argument for the settlement is partial. The complaint's investment-adequacy theory, that the VRR cap is too high given near-term entry expectations, is strongest for greenfield generation (36–48 month deployment) and weakest for demand response, storage, and existing-plant upgrades (6–18 months). For the 2026/27 BRA held 10.2 months ahead, at least some capacity categories could have responded to price signals. More fundamentally, prices inform investment even when direct construction is infeasible within the lead time: through retirement deterrence, mobile resource deployment, portfolio contracting, regulatory signaling, and the formation of expectations for future delivery years (Section 8.3). The correct structural remedy for lead-time compression is schedule restoration, which PJM is pursuing, not price suppression.

Fifth, structural alternatives dominate price controls. Three interventions reduce SFE equilibrium prices without suppressing the scarcity signal: CETL expansion (30% fringe increase $\rightarrow -24$ pp price-cost margin reduction; 10 pp CETL/reliability-requirement increase $\rightarrow$ 4–6 pp reduction); VRR slope steepening beyond the ER25-1357 baseline (slope factor ≥ 1.4 moves 2026/27 from cap-binding to interior clearing even at $K = 3$); and entry promotion to reach $K \geq 4$ (which alone would reduce 2026/27 prices by \$101/MW-day without any demand-curve modification). Each of these options is permanent and preserves the price signal, whereas the settlement is temporary and suppresses it (Section 10).

The central finding is that PJM's 2026/27 and 2027/28 capacity prices are high in a manner consistent with structurally thin supply and tight transmission constraints: three sellers hold pivotal capacity in every benchmark year, and CETL limits restrict competitive imports in the most constrained LDAs. The settlement addresses the symptom (high prices) rather than the cause (concentrated supply with limited competitive fringe). It is defensible as emergency relief during a period of acute auction-schedule disruption. It is not a substitute for the transmission investment, VRR redesign, and entry promotion that would make a similar intervention unnecessary in future auctions.

Several directions for future work follow naturally from the model's limitations. An asymmetric-firm SFE would characterize how cost and capacity heterogeneity interact with the TPS screen and the settlement's LDA application. A dynamic model with endogenous entry and exit would connect the short-run price-cost margin findings to long-run capacity adequacy under the price band. An empirical structural model recovering firm-level cost parameters from observed offer data would allow direct welfare calculation and would permit testing whether the $K = 3$ symmetric calibration understates or overstates actual strategic markups. The calibrated SFE results in this paper provide a structural benchmark for each of these extensions and a quantitative foundation for evaluating the settlement's long-run consequences as the 2028/29 auction cycle approaches.

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